

Playbook: Automated Continuous Testing for Defense-Unique LLM Compliance

This playbook outlines a modern, automated approach to streamlining the compliance pipeline for Large Language Models (LLMs) developed for or deployed within Department of Defense (DoD) environments. It addresses the "friction points" of traditional procurement by shifting to a continuous **Test, Evaluation, Verification, and Validation (TEVV)** framework.

Phase 1: Doctrinally Driven Requirement Specification

Standard AI compliance often fails because it lacks the legal and ethical nuances of military operations.

- **Automated Requirement Extraction:** Utilize internal tools like **NIPRGPT** to ingest primary source doctrine—including the **Law of War**, **Rules of Engagement (ROE)**, and **Joint Ethics Regulation (JER)**—to generate task-specific compliance benchmarks.
 - **Digital Engineering Integration:** Align requirements with the DoD Digital Engineering Strategy, using GenAI to automate the specification and analysis steps of requirements engineering.
 - **Establish Baseline Metrics:** Define metrics for accuracy, refusal rates (to avoid "over-alignment" or false refusals of lawful orders), and perplexity.
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Phase 2: Automated TEVV Infrastructure (CI/CD Integration)

To move at the speed of mission, testing must be embedded directly into the software development lifecycle.

- **CI/CD Pipeline Embedding:** Integrate evaluation tools (e.g., **DeepEval**, **RAGAS**) into the deployment pipeline to automatically trigger testing upon any model update or data retraining.
 - **LLM-as-a-Judge Framework:** Deploy a secondary, lightweight LLM to evaluate the primary model's outputs against **Key Decision-Making Attributes (KDMAs)**, providing reasoning statements for compliance auditing.
 - **Synthetic Data Generation:** Use a "Model-in-the-Loop" pipeline to transform raw doctrinal clauses into structured multiple-choice questions for scalable, diverse benchmarking.
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Phase 3: Specialized Defense Evaluation Benchmarks

Generic social safety benchmarks (e.g., toxicity) are insufficient for defense use cases.

- **Deploy ARMOR 2025:** Utilize the **ARMOR 2025** benchmark to evaluate the LLM's ability to handle the cognitive demands of the **Observe-Orient-Decide-Act (OODA)** framework.

- **Operational Refusal Testing:** Specifically test for the **False Refusal Rate (FRR)** to ensure the model does not refuse lawful, mission-critical requests due to over-sensitive commercial safety filters.
- **Edge Case Simulation:** Use simulation engines (e.g., Pulse Physiology Engine for medical triage) to generate synthetic, high-stakes scenarios for testing algorithmic decision-makers.

Phase 4: Continuous Cybersecurity & Adversarial Monitoring

AI introduces unique attack vectors that require real-time automated defense.

- **Automated Red-Teaming:** Implement continuous, crowdsourced, or tool-based red-teaming to identify vulnerabilities such as **prompt injection**, **jailbreaking**, and **model poisoning**.
- **Real-Time Guardrails:** Deploy programmable guardrails (e.g., **NVIDIA NeMo Guardrails**, **LLM Guard**) to intercept and sanitize inputs/outputs before they reach the user or secure network.
- **Outlier & Drift Detection:** Use libraries like **Alibi Detect** or **PyOD** to automatically identify concept drift or out-of-distribution inputs that could degrade model performance over time.

Phase 5: Streamlined Compliance & "Assess Only" ATO Pathways

Traditional Authority to Operate (ATO) processes take months; continuous software requires a faster path.

- **The "Assess Only" Construct:** For models deployed on pre-authorized hardware/cloud infrastructure, utilize the DoD's **"Assess Only"** process. This allows teams to submit evidence (e.g., model cards, TEVV reports) directly to the existing infrastructure's security package, bypassing a full hardware-level RMF review.
- **Automated Documentation:** Dynamically update **Data Cards** and **Model Cards** using tools that track data provenance, training algorithms, and evaluation results to provide a "living" compliance record.
- **Incident Repositories:** Report failures and "near-misses" to the central **CDAO AI Incident Repository** to inform future risk mitigations across the DoD ecosystem.

Recommended Toolbox for Continuous Testing

Capability

Recommended Tools/Datasets

Military Alignment	ARMOR 2025, GAMECHANGER Policy Database
RAG Evaluation	RAGAS, HotpotQA, DeepEval
Robustness/Adversarial	AdversarialGLUE, Prompt Fuzzer, CounterFit
Observability/Tracing	Arize Phoenix, MLflow
Safety Guardrails	NVIDIA NeMo Guardrails, LLM Guard
PII/Privacy	Microsoft Presidio, IBM Data Privacy Toolkit